

Concursul de matematică “Ștefan Musta”

Ediția XXX

05 aprilie 2025

Clasa a VII-a

Barem de corectare

PROBLEMA 1

$$x = \sqrt{15a + 15b} \dots\dots\dots 1p$$

$$x = \sqrt{15(a + b)}, x \in \mathbb{N} \text{ și } 15(a + b) \text{ p.p.} \dots\dots\dots 2p$$

$$a + b = 15 \dots\dots\dots 2p$$

$$\overline{ab} \in \{69, 78, 87, 96\} \dots\dots\dots 2p$$

PROBLEMA 2

$$a) (a + 2)^2 + (b - 1)^2 = 0 \Leftrightarrow a = -2, b = 1 \dots\dots\dots 2p$$

$$(a + b)^{100} + b^{100} = 2 \dots\dots\dots 1p$$

$$b) \frac{x}{y} + 1 = 2 + \sqrt{2} \dots\dots\dots 1p$$

$$\frac{x}{y} = 1 + \sqrt{2} \dots\dots\dots 1p$$

$$\frac{x-y}{y} = \frac{x}{y} - 1 = 1 + \sqrt{2} - 1 = \sqrt{2} \dots\dots\dots 2p$$

PROBLEMA 3

PN – linie mijlocie în ΔAMC

$$PN = \frac{MC}{2} = \frac{BC}{4} \dots\dots\dots 2p$$

$$\text{Fie } AD \perp BC, D \in BC, AD \cap PN = \{R\} \dots\dots\dots 1p$$

$$A_{\Delta APN} = \frac{PN \cdot AR}{2} = \frac{\frac{BC}{4} \cdot \frac{AD}{2}}{2} = \frac{BC \cdot AD}{16} \dots\dots\dots 2p$$

$$A_{\Delta APN} = \frac{A_{\Delta ABC}}{8} = 0,125 A_{\Delta ABC} \dots\dots\dots 1p$$

$$A_{\Delta APN} = 12,5\% \cdot A_{\Delta ABC} \dots\dots\dots 1p$$